

Advanced Deep Learning — Final Project: AI Automation

Overview

You will design and build a working AI automation system on a topic of your choice, using a low-code automation platform (n8n, Zapier, Make, or equivalent) as the backbone. The system must combine the techniques covered in the course: prompt engineering, RAG, multimodality, and agentic AI.

The topic is free. Pick a real, useful use case — an internal support assistant, a document processing pipeline, a media monitoring tool, a recruitment screener, whatever you find relevant. The topic itself is not graded; the engineering choices behind it are.

Organisation

- Teams of 5 to 6 students.
- Duration: 5 weeks.
- A progress check-in takes place every week during the course session. Attendance and a short update are mandatory for every team.
- Final defence: last session of the course.

Mandatory requirements

1. Automation platform The pipeline must run on an automation tool (n8n, Zapier, Make, etc.). Everything should be triggerable end-to-end, not run manually cell by cell.

2. Prompt engineering This is the core of the evaluation. You must keep a versioned prompt library: for each prompt, document its role in the system, the technique used (few-shot, chain-of-thought, structured output, role prompting, guardrails...), the failure cases you observed, and how you fixed them. Show at least one before/after iteration per key prompt.

3. RAG You must build your own knowledge base. Justify:

- the choice of the corpus (source, size, format, why it fits the use case);
- the chunking strategy (fixed size, semantic, recursive, by document structure...), the chunk size and overlap, and why;
- the embedding model and the vector store;
- the retrieval setup (top-k, filters, reranking if any). Provide a short evaluation of retrieval quality on a small set of test queries. A comparison of two chunking strategies is strongly valued.

4. Multi-agent system Your system must involve several agents, not a single monolithic call. You must justify the architecture you chose (orchestrator, router, master-worker, sequential chain, hierarchical, etc.): what each agent does, what tools it has access to, how information flows between them, and why an alternative architecture would have been worse for your case.

5. Multimodality — mandatory The system must process or generate at least one non-textual modality (image, audio, video, PDF with layout, speech). Explain how it is integrated into the pipeline and what it brings to the use case.

Deliverables

Due the day before the final session:

1. The exported workflow (e.g. n8n JSON) or repository, with instructions to run it.
2. A technical report, 8 pages maximum, covering the five points above.
3. The prompt library, as a separate file.
4. A short retrieval evaluation (test queries, results, comments).

Final session: 15-minute live demo + 10 minutes of questions. Every team member must speak and must be able to answer questions on any part of the system.

Weekly milestones

Week	Expected at the check-in
1	Use case defined and scoped, team roles, automation tool chosen, first draft of the architecture
2	Corpus collected, chunking strategy implemented, first retrieval tests
3	Agents implemented and communicating, prompts v1, multimodal component working in isolation
4	Full end-to-end integration, prompt iteration based on observed failures, error handling
5	Testing, report, demo rehearsal

Grading

Criterion	Weight
Prompt engineering (quality, iteration, documentation)	20%
RAG (corpus, chunking, retrieval, evaluation)	25%
Multi-agent architecture and its justification	25%
Multimodality integration	15%
Demo, report and Q&A	15%

Weekly check-ins are not graded separately, but a missed or empty check-in will be penalised on the final mark.

Notes

- Justification matters more than complexity. A simple architecture that you can defend beats a complicated one you cannot explain.
- Reporting what did not work, and why, earns credit. Do not hide failed attempts.
- You may use any LLM provider, including local models. Watch your API costs.